

Common Fund Data Ecosystem (CFDE) Training Center

Decoding the Data Ecosystem Institute
(DeCoDE Institute)

Part 1: Overview of CFDE & Scavenger Hunt

June 2, 2026



Welcome/Agenda

- CFDE Overview
- CFDE Training Center Overview
- DeCoDE Institute Overview
- CFDE Scavenger Hunt
- Q&A



CFDE Overview

NIH Common Fund Data Ecosystem (CFDE)

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Decoding the Data Ecosystem Institute (DeCoDE Institute)
June 2, 2026

NIH CFDE Team



George Papanicolaou,
Program Leader



Chris Kinsinger, CDR
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Manager



Laurel Kuxhaus,
Program Officer



Sahana Kukke, Program
Leader



Renee Collins, Program
Specialist



Haluk Resat, Program
Leader



David Dzamashvili,
Operations (Trio)



Natalie Vineyard, SPEC
(Trio)



Alexandria Burns,
DOTM (Trio)



Common Fund Programs

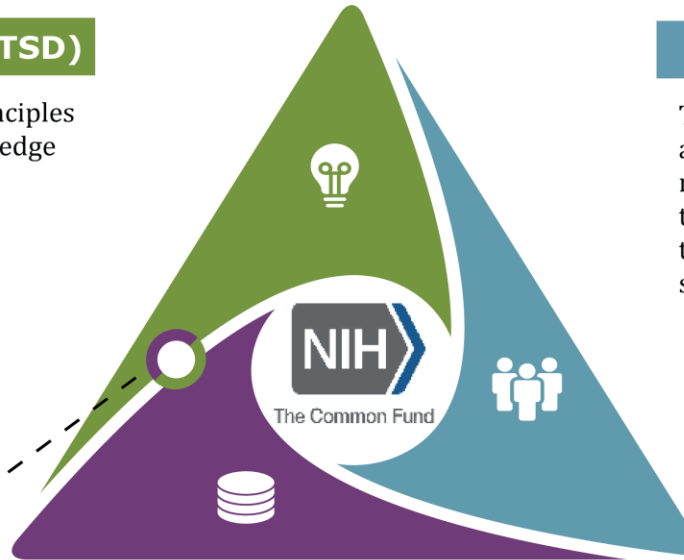
Transformational Science and Discovery (TSD)

These **programs** are designed to establish new scientific principles, models, and research resources to transform scientific knowledge and paths to discovery.

- 4D Nucleome
- A2CPS
- CryoEM
- DS-I Africa
- SMaHT
- Human Virome
- HRHR
- MoTrPAC
- NIOI
- SPARC
- NPH
- Oculomics
- SenNet
- RNomics

TSD programs that create catalytic data resources

4D Nucleome • A2CPS • Human Virome
MoTrPAC • NPH • SenNet • SMaHT • SPARC



Catalytic Data Resources (CDR)

These **programs** are designed to manage and develop data for scientific discoveries by accelerating research through data resources.

- Bridge2AI
- CFDE
- HuBMAP
- Kids First
- PRIMED-AI
- SysBio

Re-Engineering the Research Enterprise (RRE)

These **programs** are designed to transform how we do biomedical and behavioral research, how we make the biomedical workforce as robust as possible to ensure new perspectives and ideas contribute to discovery, how we transition that research into prevention and therapies, and how those successful prevention and therapies can be shared broadly.

- CARE for Health™
- Complement-ARIE
- NBSxWGS
- Replication Initiative
- SCGE
- Transformative Research to Address Health Disparities

Islands of Data

Omics

Imaging

Assays

Microscopy

Clinical

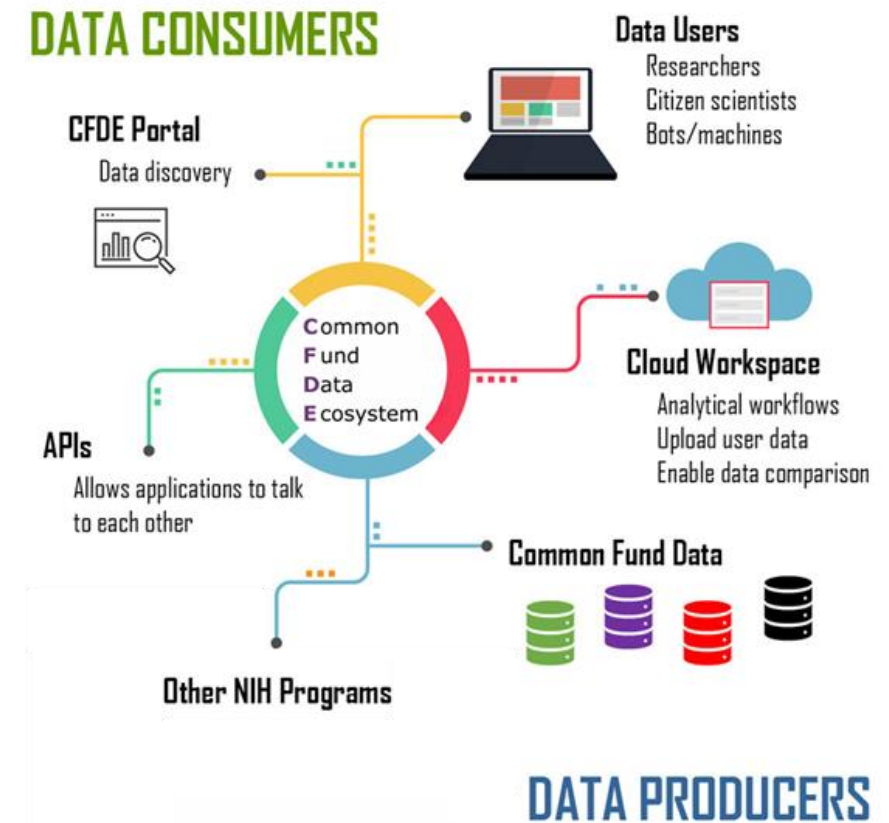


Common Fund Data Ecosystem (CFDE)

Mission: Foster scientific discovery through the (re)use of data and resources generated by Common Fund (CF) Programs

Aims:

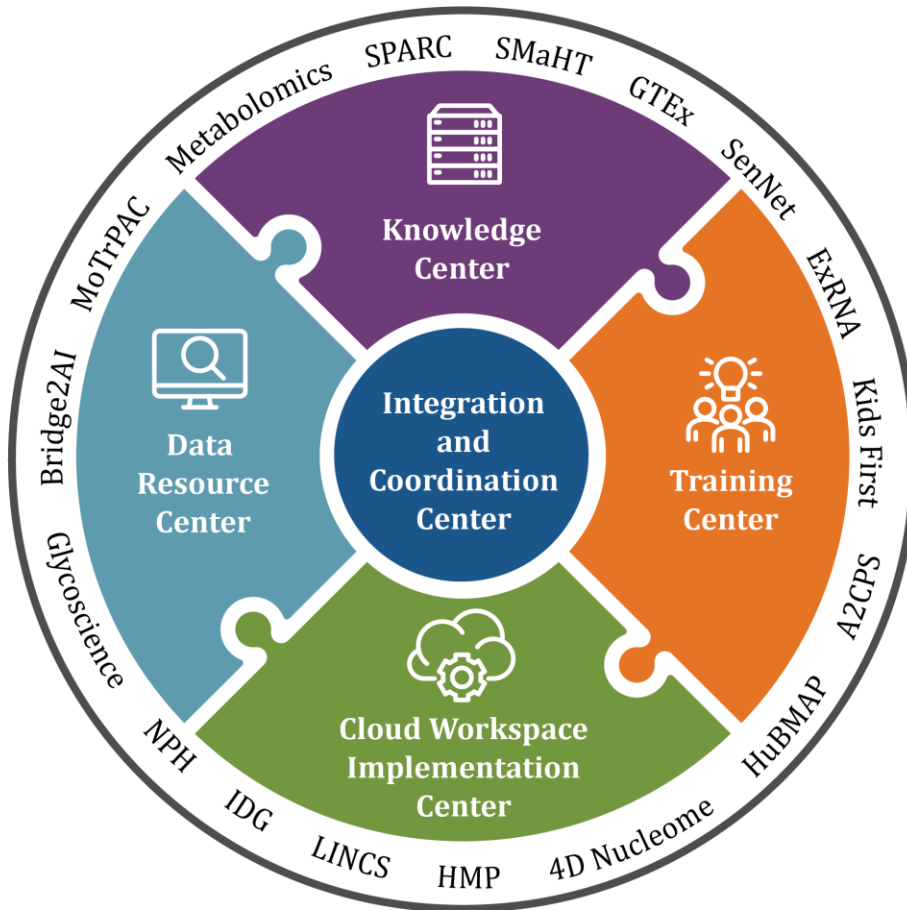
1. Enable users to **query across & use** multiple CF data sets
2. Provide **training and outreach** to bring people to CF data and train them to work in the cloud
3. **Coordinate and integrate** infrastructure and activities into a cohesive ecosystem



The chief metric of success for CFDE is discovery: If the CFDE is successful, investigators will be using Common Fund data (and products) for new discoveries and new purposes.

CFDE Approach

Core infrastructure built by five distinct but interconnected Centers



Design Pillars

1. Build resources that are complimentary, not duplicative

- No data repository, federated data storage
- Use of APIs and standards to connect with other NIH data ecosystems

2. Community-driven ecosystem, not a platform

- Community-driven and maintained
- Diversity of data and user communities means there won't be a "one-size fits all" solution

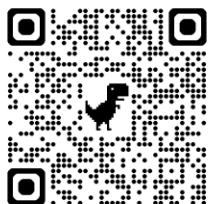
3. Scalable and sustainable

- Flexible enough to evolve with the field

Discover and understand Common Fund data



CFDE WORKBENCH



<https://data.cfde.cloud>

CFDE DATA PORTAL

Search Common Fund Programs' Metadata and Processed Datasets.

Enter one or more keywords



Try STAT3, blood, dexamethasone

LEARN MORE

GRAPH SEARCH

NEW

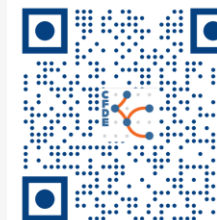
SEARCH →



THE KNOWLEDGE CENTER integrates data and knowledge generated by ground-breaking research programs in the NIH Common Fund Data Ecosystem.

SEARCH COMMON FUND KNOWLEDGE

Click here to begin



<https://cfdeknowledge.org>



National Institutes of Health
Office of Strategic Coordination–The Common Fund

Learn about and work with Common Fund data



<https://orau.org/cfde-trainingcenter/>



CFDE CLOUD WORKSPACE

<https://cfdeworkspace.org/>



CFDE Training Center Overview

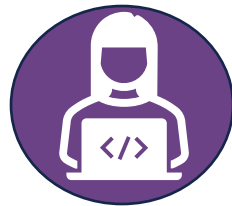
CFDE Training Center (TC) Overview

The TC provides training in basic and advanced computational and data analytics skills for data science learners and users interested in CFDE data, tools, and resources.

The TC aims to expand the CFDE data userbase and enhance the confidence and complexity of dataset usage through community building and engagement activities.



**CFDE TC
Podcast**



**e-Learning
Dashboard**



**Hackathons and
Workshops**



**DeCoDE
Institute**



**Education &
Engagement**



DeCoDE Institute Learning Objectives

- Develop skills to analyze omics datasets and build workflows using CFDE tools and resources.
- Engage in collaborative, hands-on learning experiences and foster innovation through live training sessions.
- Utilize CFDE resources and e-learning modules for flexible, asynchronous learning opportunities.
- Demonstrate knowledge and skills by participating in team-based capstone projects.

Coming Soon from the



- June 11: Speed Networking
- June 15 & 17: Part 1 & 2:
Applied AI for Biologists Using
Galaxy and CFDE Cloud
Resources



Submit by June 30 11:59 p.m. PT

Use data from one or more CFDE-supported programs to create a visualization that tells a compelling biomedical story. Up to **\$1000 in prizes** will be awarded to winning visualizations!

Learn more and register: www.oraui.org/cfde-trainingcenter

And more ...

- **Monthly Podcast** – *Decoding the Data Ecosystem* is dedicated to unraveling the complexities and exploring the depths of omics research
- **e-Learning Dashboard** – Free, innovative omics training on-demand and home to the educational 'How-To Seminar Series'
- Follow us on Linked  **@cfde-trainingcenter** to stay informed of upcoming training and events!

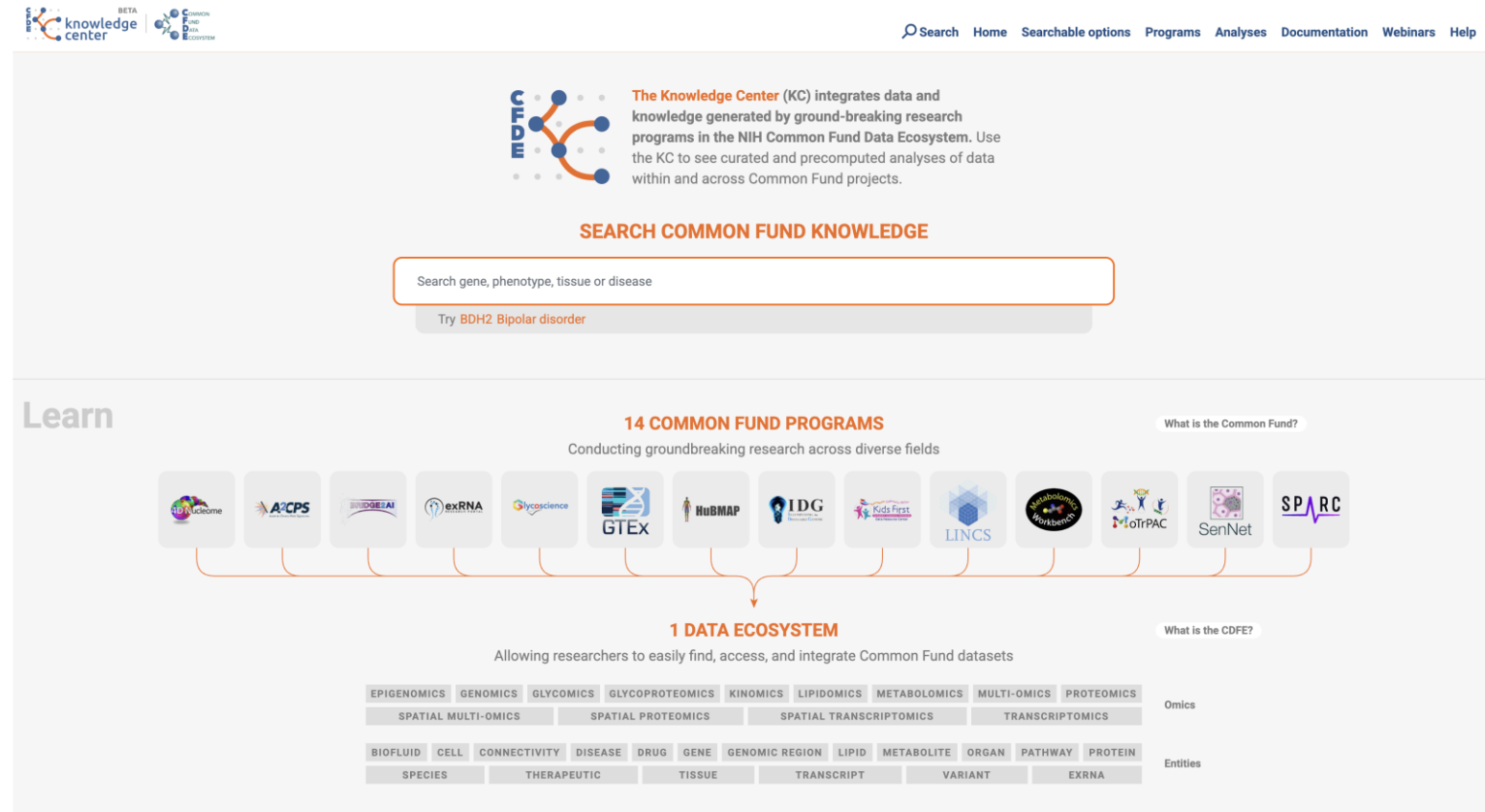


CFDE Scavenger Hunt

Now, to the activity!

- Learn to navigate the various CFDE pages:
 1. <https://commonfund.nih.gov/dataecosystem>
 - Information about CFDE as a whole and a snapshot of the different funded programs
 2. https://cfdeknowledge.org/r/kc_landing
 - Information about the kinds of data in each program as well as curated and precomputed analysis of data
 3. <https://data.cfde.cloud/>
 - CFDE workbench tools, chatbot, workflows, use cases, and more
- Participate in the Scavenger Hunt:
 - Learn more about the CFDE programs and types of data found in each
 - Find and use specific CFDE tools and data
 - Identify new information for potential follow-up explorations

Who is participating in CFDE?



The screenshot shows the CFDE Knowledge Center landing page. At the top, there's a navigation bar with links: Search, Home, Searchable options, Programs, Analyses, Documentation, Webinars, and Help. Below the navigation bar, the page features a large header section with the CFDE logo and a description: "The Knowledge Center (KC) integrates data and knowledge generated by ground-breaking research programs in the NIH Common Fund Data Ecosystem. Use the KC to see curated and precomputed analyses of data within and across Common Fund projects." Below this is a search bar with the placeholder text "Search gene, phenotype, tissue or disease" and a suggestion "Try BDH2 Bipolar disorder".

The main content area is titled "Learn" and is divided into two sections. The first section is "14 COMMON FUND PROGRAMS" with the subtitle "Conducting groundbreaking research across diverse fields". It lists 14 programs: 10 Nucleome, A2CPS, BRIDGE AI, exRNA, Glycoscience, GTEx, HuBMAP, IDG, Kids First, LINCS, MIND, MoTrPAC, SenNet, and SPARC. The second section is "1 DATA ECOSYSTEM" with the subtitle "Allowing researchers to easily find, access, and integrate Common Fund datasets". It lists various data types: EPIGENOMICS, GENOMICS, GLYCOMICS, GLYCOPROTEOMICS, KINOMICS, LIPIDOMICS, METABOLOMICS, MULTI-OMICS, and PROTEOMICS. Below these are two rows of entities: BIOFLUID, CELL, CONNECTIVITY, DISEASE, DRUG, GENE, GENOMIC REGION, LIPID, METABOLITE, ORGAN, PATHWAY, and PROTEIN. The bottom row lists SPECIES, THERAPEUTIC, TISSUE, TRANSCRIPT, VARIANT, and EXRNA.

https://cfdeknowledge.org/r/kc_landing

What types of data are contributed?

CFDE Programs

**4DN**
4D Nucleome (4DN)
<https://www.4dnucleome.org/>

Description The 4D Nucleome program (4DN) aims to understand the 3D organization of DNA in the nucleus, how it varies over time and in different cell types, and its role in health and disease.

Omics Epigenomics Genomics Transcriptomics

Entities Cell Disease Genomic Region Species

Data Resources Explore 4DN datasets through the [4D Nucleome Data Portal](#) or the [CFDE Workbench](#).

More >

**A2CPS**
Acute to Chronic Pain Signatures (A2CPS)
<https://a2cps.org/>

Description The Acute to Chronic Pain Signatures (A2CPS) project seeks to identify biomarkers and signatures of risk and resilience to the transition to chronic pain after surgery.

Omics Genomics Lipidomics Metabolomics Proteomics Transcriptomics

Entities Connectivity Gene Lipid Metabolite Protein Transcript

Data Resources A2CPS is currently generating data; stay tuned for results.

More >

Filter data

Omics

- ☐ Check / Uncheck all
- ☒ Epigenomics
- ☐ Genomics
- ☐ Glycomics
- ☐ Glycoproteomics
- ☐ Kinomics
- ☐ Lipidomics
- ☐ Metabolomics
- ☐ Multi-omics
- ☐ Proteomics
- ☐ Spatial Multi-omics
- ☐ Spatial Proteomics
- ☐ Spatial Transcriptomics
- ☐ Transcriptomics

Entities

- ☐ Check / Uncheck all
- ☐ Biofluid
- ☐ Cell
- ☐ Connectivity
- ☐ Disease
- ☐ Drug

https://cfdeknowledge.org/r/kc_programs

[CFDE-REVEAL](#)

[CFDE-DESIGN](#)

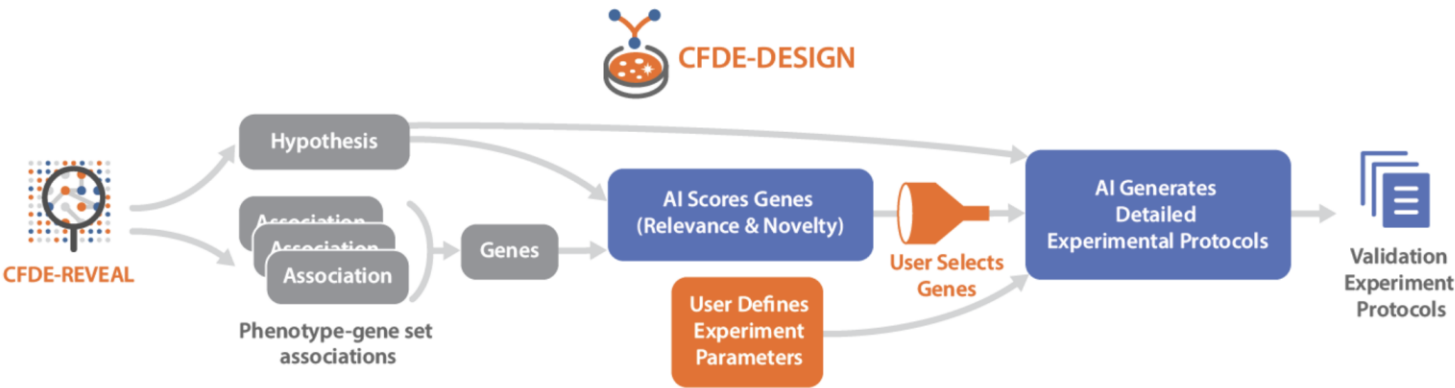
[4DN Falcon Analysis](#)

[Differential Expression Analysis](#)

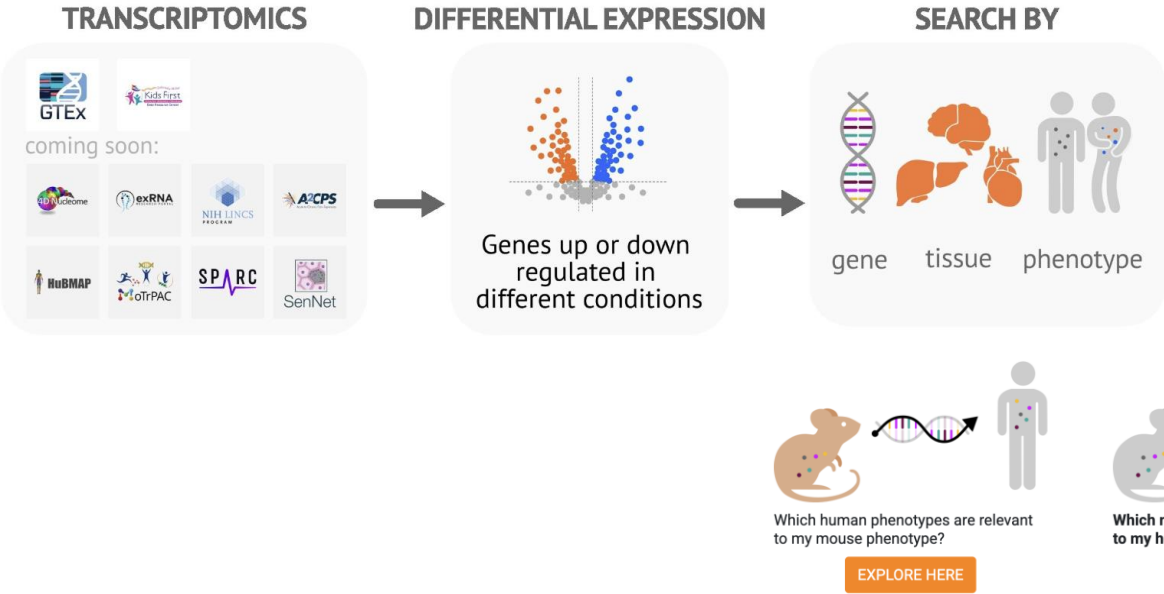
[Gene Set Analysis](#)

[Mouse to Human](#)

How can people interact with the data?



Differential Expression | All Results and Gene Sets



<https://cfdeknowledge.org>

How can people interact with the data?

CFDE WORKBENCH

DATA CLOUD KNOWLEDGE TRAINING COORDINATION LOGIN

SEARCH EXPLORE THE ECOSYSTEM ANALYZE ENGAGE SUBMIT ABOUT

Discover, Analyze, and Integrate NIH Common Fund Data

The Common Fund Data Ecosystem (CFDE) aims to enable broad use of data generated by Common Fund Programs to accelerate hypothesis generation and discovery.

Enter a biomedical entity to get started

Small Molecules, Drugs, and Metabolites
Enter small molecules, drugs, or metabolit...

Gene Sets, Pathways, and Modules
Search gene sets

Genes, Proteins, and Variants
Enter genes, proteins, or variants name

Cell types, Tissues, and Organ
Enter cell types, tissues, or organ name

Phenotypes, and Diseases
Enter phenotypes, or diseases name

Assay
Enter assay name

10,239,060 Files 1,256,461 Knowledge Graph Assertions 362,157 Compounds 56,844 Gene Sets 34,193 Genes Click To see more

<https://cfde.cloud/>

How can people interact with the data?



CWIC
Portal



Command
Line



Interactive



TACC



Data
Services



High Performance
Computing



Cloud
Computing

<https://cfdeworkspace.org/>

What are researchers currently doing?



CFDE WORKBENCH

[INFORMATION PORTAL](#)

[DATA PORTAL](#)

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[INFO](#) > [PUBLICATIONS](#)

CFDE Associated and Common Fund Programs' Landmark Publications

☒ Centers ☒ DCCs ☒ Landmark ☒ R03s

Diwan R, Gaytan SL, Bhatt HN, Pena-Zacarias J, Nurunnabi M. 2024. Liver fibrosis pathologies and potentials of RNA based therapeutics modalities. *Drug Delivery And Translational Research*. 14(10):2743-2770. <https://www.doi.org/10.1007/s13346-024-01551-8>

[PubMed](#)

[DOI](#)

[R03](#)

Animals

Liver Cirrhosis

Rna

Lipid Nanoparticle

Rna Therapeutics

Mrna

Nanoparticles

Drug Delivery Systems

Sirna

Humans

Liver Fibrosis

Beaven E, Kumar R, An JM, Mendoza H, Sutradhar SC, Choi W, Narayan M, Lee YK, Nurunnabi M. 2024. Potentials of ionic liquids to overcome physical and biological barriers. *Advanced Drug Delivery Reviews*. 204:115157. <https://www.doi.org/10.1016/j.addr.2023.115157>

[PubMed](#)

[PMC](#)

[DOI](#)

[Export Citation](#)

[R03](#)

Administration, Cutaneous

Biological Barrier

Ionic Liquid

Ionic Liquids

Local Delivery

Noninvasive Delivery

Drug Delivery Systems

Systemic Delivery

Humans

Fisher JL, Wilk EJ, Oza VH, Gary SE, Howton TC, Flanary VL, Clark AD, Hjelmeland AB, Lasseigne BN. 2024. Signature reversion of three disease-associated gene signatures prioritizes cancer drug repurposing candidates. *Febs Open Bio*. 14(5):803-830. <https://www.doi.org/10.1002/2211-5463.13796>

[PubMed](#)

[PMC](#)

[DOI](#)

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[R03](#)

Gene Expression Profiling

Cancer

Drug Repositioning

Glioblastoma

Animals

Glioblastoma

Gene Expression Regulation, Neoplastic

Gene Signature

Mice

Brain Neoplasms

Transcriptomic Signature

Xenograft Model Antitumor Assays

Transcriptome

Antineoplastic Agents

Neoplasms

Cell Line, Tumor

Drug Repurposing

Humans

Gjoni K, Pollard KS. 2024. SuPreMo: a computational tool for streamlining in silico perturbation using sequence-based predictive models. *Bioinformatics (oxford, England)*. 40(6). <https://www.doi.org/10.1093/bioinformatics/btae340>

[PubMed](#)

[PMC](#)

[DOI](#)

[Export Citation](#)

[R03](#)

<https://info.cfde.cloud/publications>

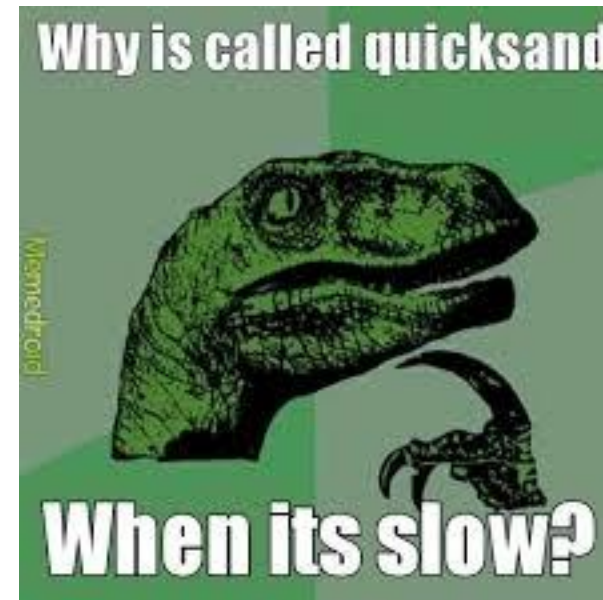


CFDE
TRAINING
CENTER

Instructions

- Participants will be assigned to teams of two
- Teams will have ~30 minutes to work through the Digital Scavenger Hunt
 - The last question is open-ended: asking participants to find an interesting observation from their favorite dataset
- Answers will be reviewed together as a group with demonstrations
 - A different team will get called on to answer each question
 - There will be an opportunity to share the interesting observations
- General Q&A
- https://bit.ly/Scavenger_Hunt_template

Q & A Time!



Contact Us



Learn more about the Training Center:
<https://orau.org/cfde-trainingcenter>

Connect with us on [LinkedIn](#)
<https://www.linkedin.com/company/cfde-trainingcenter>

Questions? Interested in collaborating or being a podcast guest?
Email us: cfde-trainingcenter@orau.org

Please Take Our Survey



<https://orausurvey.orau.org/n/DeCoDEscavengerhunt.aspx>